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1. 习题二:

30.

$$(1) P(X > 1) = \int_1^{+\infty} f(x)dx = \int_1^2 \frac{1}{9}(4 - x^2)dx = \frac{5}{27}$$

故化学反应发生的概率为 $\frac{5}{27}$

(2) 由题意得, Y 符合二项分布, Y 的分布律为

$$P(Y = k) = C_{10}^k \left(\frac{5}{27}\right)^k \left(\frac{22}{27}\right)^{10-k}, k = 1, 2, \dots, 10$$

(3) 由 (2) 知

$$P(Y = 2) = C_{10}^2 \left(\frac{5}{27}\right)^2 \left(\frac{22}{27}\right)^8 \approx 0.2998$$

$$P(Y \geq 2) = 1 - P(Y < 2) = 1 - C_{10}^0 \left(\frac{22}{27}\right)^{10} - C_{10}^1 \left(\frac{5}{27}\right) \left(\frac{22}{27}\right)^9 \approx 0.578$$

32.

(1) 令 A = “厂家免费为其更换显示器”, X 的分布函数为 $F_X(x) = 1 - e^{-\frac{x}{50}}, (x > 0)$

$$P(A) = P(X \leq 2) = F_X(2) = 1 - e^{-0.04} \approx 0.0392$$

(2) 令 B = “显示器至少可以使用 10000h”

$$P(B) = P(X > 10) = 1 - P(X \leq 10) = 1 - F_X(10) = e^{-0.2} \approx 0.8187$$

(3) 令 C = “显示器至少可以使用 20000h”

$$P(C|B) = \frac{P(BC)}{P(B)} = \frac{P(C)}{P(B)} = \frac{P(X > 20)}{P(B)} = \frac{1 - F_X(20)}{P(B)} = e^{-0.2} \approx 0.8187$$

37.

令 A = “螺栓不合格”

$$P(A) = 1 - P(9.88 < X < 10.12) = 1 - 0.9544 = 0.0456$$

2 39.

令 A = “在一次测量中误差的绝对值大于 23.26”, B = “至少有四次测量误差的绝对值大于 23.26”

$$P(A) = 1 - P(|X| \leq 23.26) = 1 - (2\Phi(\frac{23.26}{10}) - 1) = 0.02$$

$$P(B) = 1 - \sum_{k=0}^3 C_{200}^k 0.02^k 0.98^{200-k} \approx 1 - \sum_{k=0}^3 \frac{4^k}{k!} e^{-4} \approx 0.5665$$

41.

由题意得 X 为离散型随机变量, X 的分布律为

X	-2	-1	2
P	0.3	0.6	0.1

故 Y 的可能取值为 1, -2

$$P(Y = 1) = P(X = 2) + P(X = -2) = 0.4, P(Y = -2) = P(X = -1) = 0.6$$

Y 的分布律为

Y	-2	1
P	0.6	0.4

故 Z 的可能取值为 1, 2

$$P(Z = 2) = P(X = 2) + P(X = -2) = 0.4, P(Z = 1) = P(X = -1) = 0.6$$

Z 的分布律为

Y	1	2
P	0.6	0.4

42.

$$(1) Y_1 \text{ 的可能取值为 } -1, 1, x \text{ 的密度函数为 } f(x) = \begin{cases} 0.2, & -2 < x < 3 \\ 0, & x \notin (-2, 3) \end{cases}$$

$$P(Y_1 = -1) = P(X < 0) = \int_{-2}^0 f(x) dx = 0.4$$

$$P(Y_1 = 1) = 1 - P(Y_1 = -1) = 0.6$$

Y_1 的分布律为

Y	-1	1
P	0.4	0.6

(2) Y_2 的分布函数为

$$F_{Y_2}(y) = P(Y_2 \leq y) = P\left(\frac{X+1}{2} \leq y\right) = P(X \leq 2y-1) = F_X(2y-1) = \begin{cases} 0, & y < -0.5 \\ \frac{2y+1}{5}, & -0.5 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$

Y_2 的概率密度为

$$f_{Y_2}(y) = \frac{d}{dy}F_{Y_2}(y) = \begin{cases} 0.4, & -0.5 \leq y \leq 1 \\ 0, & \text{else} \end{cases}$$

43.

(1) Y_1 的分布函数为

$$F_{Y_1}(y) = P(Y_1 \leq y) = P(X^2 \leq y) = \begin{cases} \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) = \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{y}}^{\sqrt{y}} e^{-\frac{s^2}{2}} ds, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

Y_1 的概率密度为

$$f_{Y_1}(y) = \frac{d}{dy}F_{Y_1}(y) = \begin{cases} \frac{1}{\sqrt{2\pi y}} e^{-\frac{y}{2}}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

(2) Y_2 的分布函数为

$$F_{Y_2}(y) = P(Y_2 \leq y) = P(e^{-X} \leq y) = \begin{cases} 1 - \Phi(-\ln y) = 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-\ln y} e^{-\frac{s^2}{2}} ds, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

Y_2 的概率密度为

$$f_{Y_2}(y) = \frac{d}{dy}F_{Y_2}(y) = \begin{cases} \frac{1}{\sqrt{2\pi y}} e^{-\frac{\ln^2 y}{2}}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

(3) Y_3 的分布函数为

$$F_{Y_3}(y) = P(Y_3 \leq y) = P(X + |X| \leq y) = \begin{cases} 0, & y < 0 \\ \Phi\left(\frac{y}{2}\right), & y \geq 0 \end{cases}$$

$$\text{则 } f_{Y_3}(y) = \begin{cases} 0, & y < 0 \\ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{y}{2}} e^{-\frac{s^2}{2}} ds \end{cases}$$

45.

4 设此点所落位置与圆心连线与 x 轴正方向之间的夹角为 Θ ，则 $\Theta \sim U(0, 2\pi)$

Θ 的分布函数为

$$F_{\Theta}(\theta) = \begin{cases} 0, & \theta < 0 \\ \frac{\theta}{2\pi}, & 0 \leq \theta < 2\pi \\ 1, & \theta > 2\pi \end{cases}$$

X 的分布函数为

$$F_X(x) = P(X \leq x) = P(\cos\Theta \leq x) = \begin{cases} 0, & |x| > 1 \\ F_{\Theta}(2\pi - \arccos x) - F_{\Theta}(\arccos x) = \frac{\pi - \arccos x}{\pi}, & |x| \leq 1 \end{cases}$$

X 的概率密度为

$$f(x) = \frac{d}{dy} F_X(x) = \begin{cases} \frac{1}{\pi\sqrt{1-x^2}}, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$$

开放式案例分析题：

问：公司按现金补贴股票年化收益率的概率是多少？

解：令 A = “公司按现金补贴股票年化收益率”

$$\int_{-\infty}^{+\infty} f_X(x) dx = 1$$

$$\text{解得 } \lambda = \frac{e^{2.5} - e^{-2.5}}{2}$$

$$P(A) = P(X < 3) = \int_{-\infty}^3 f(x) dx = \int_0^3 \lambda e^{-0.5(x-5)} = \frac{e^{2.5} - e}{e^{2.5} - e^{-2.5}}$$